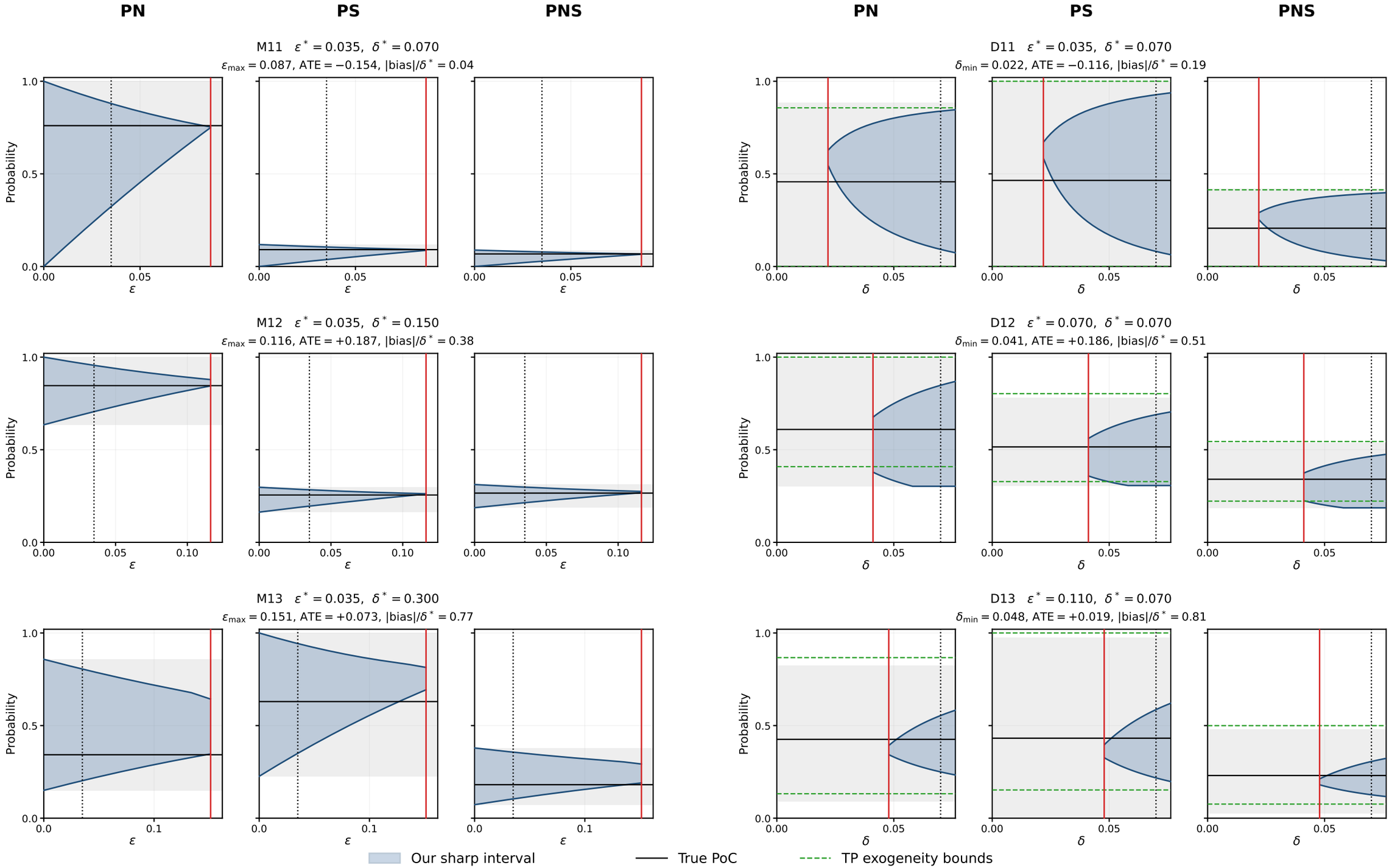
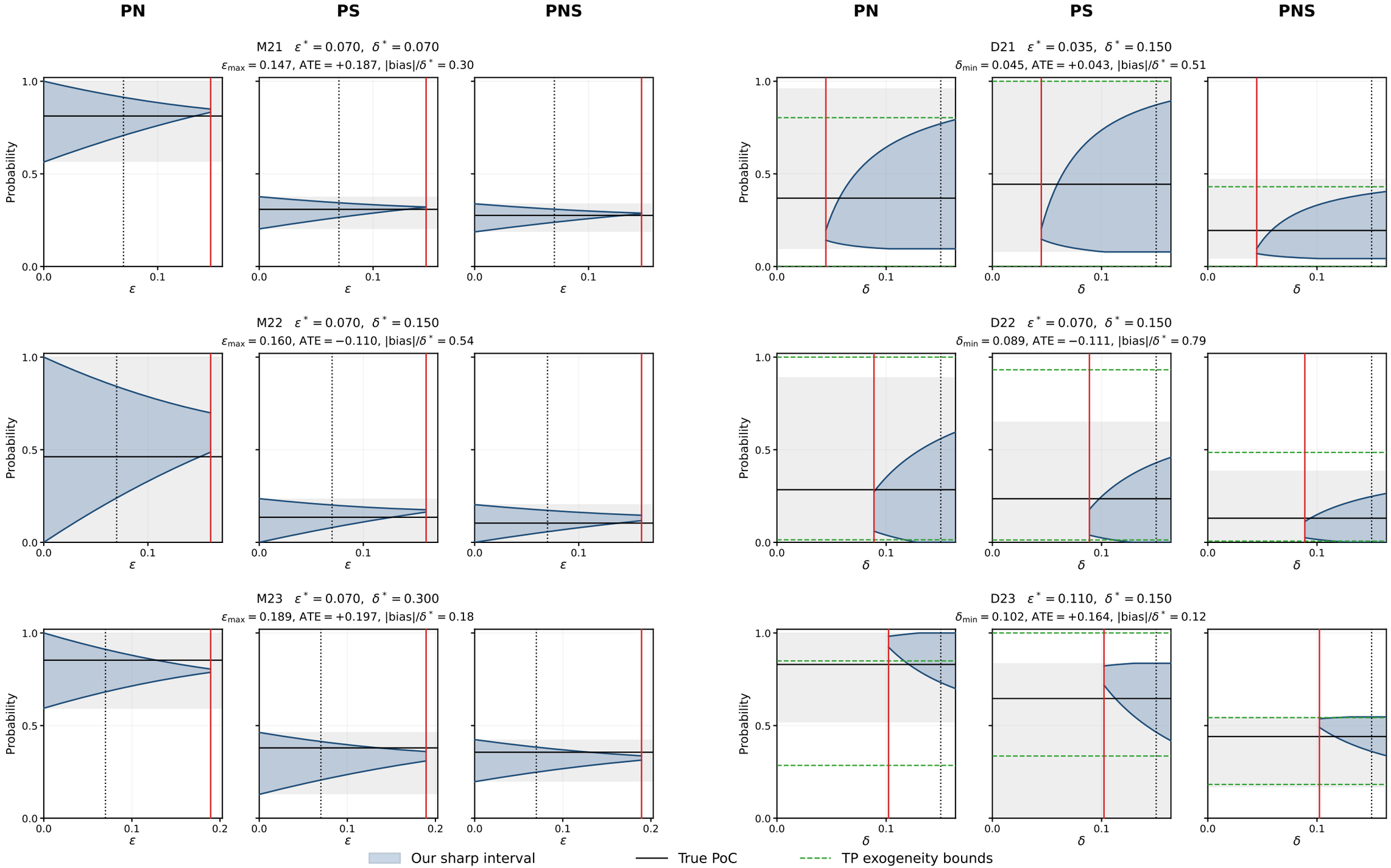
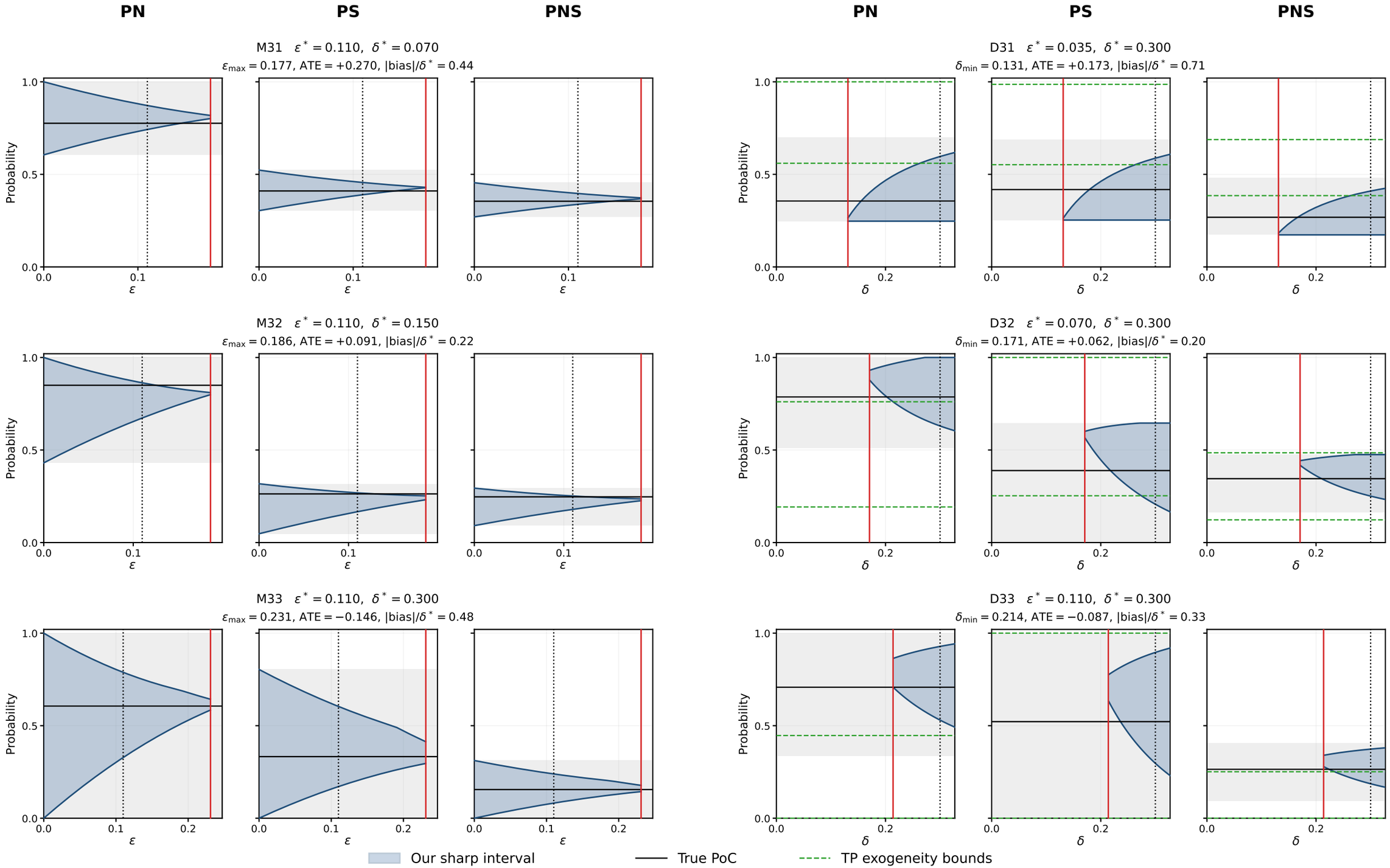

PN
PS
PNS
PN
PS
PNS
M11 $\varepsilon^* = 0.035$, $\delta^* = 0.070$
$\varepsilon_{\max} = 0.087$, ATE $= -0.154$, |bias|$/\delta^* = 0.04$
D11 $\varepsilon^* = 0.035$, $\delta^* = 0.070$
$\delta_{\min} = 0.022$, ATE $= -0.116$, |bias|$/\delta^* = 0.19$
M12 $\varepsilon^* = 0.035$, $\delta^* = 0.150$
$\varepsilon_{\max} = 0.116$, ATE $= +0.187$, |bias|$/\delta^* = 0.38$
D12 $\varepsilon^* = 0.070$, $\delta^* = 0.070$
$\delta_{\min} = 0.041$, ATE $= +0.186$, |bias|$/\delta^* = 0.51$
M13 $\varepsilon^* = 0.035$, $\delta^* = 0.300$
$\varepsilon_{\max} = 0.151$, ATE $= +0.073$, |bias|$/\delta^* = 0.77$
D13 $\varepsilon^* = 0.110$, $\delta^* = 0.070$
$\delta_{\min} = 0.048$, ATE $= +0.019$, |bias|$/\delta^* = 0.81$
Probability
ε
δ
Our sharp interval
Assumption-free TP interval
True PoC
Oracle value
TP exogeneity bounds
Compatibility threshold


PN
PS
PNS
PN
PS
PNS
M21 $\varepsilon^* = 0.070$, $\delta^* = 0.070$
$\varepsilon_{\max} = 0.147$, ATE $= +0.187$, $|\text{bias}|/\delta^* = 0.30$
D21 $\varepsilon^* = 0.035$, $\delta^* = 0.150$
$\delta_{\min} = 0.045$, ATE $= +0.043$, $|\text{bias}|/\delta^* = 0.51$
M22 $\varepsilon^* = 0.070$, $\delta^* = 0.150$
$\varepsilon_{\max} = 0.160$, ATE $= -0.110$, $|\text{bias}|/\delta^* = 0.54$
D22 $\varepsilon^* = 0.070$, $\delta^* = 0.150$
$\delta_{\min} = 0.089$, ATE $= -0.111$, $|\text{bias}|/\delta^* = 0.79$
M23 $\varepsilon^* = 0.070$, $\delta^* = 0.300$
$\varepsilon_{\max} = 0.189$, ATE $= +0.197$, $|\text{bias}|/\delta^* = 0.18$
D23 $\varepsilon^* = 0.110$, $\delta^* = 0.150$
$\delta_{\min} = 0.102$, ATE $= +0.164$, $|\text{bias}|/\delta^* = 0.12$
Probability
ε
δ
Our sharp interval
Assumption-free TP interval
True PoC
Oracle value
TP exogeneity bounds
Compatibility threshold


PN
PS
PNS
M31 $\varepsilon^* = 0.110,\ \delta^* = 0.070$
$\varepsilon_{\max} = 0.177$, ATE $= +0.270$, $|\text{bias}|/\delta^* = 0.44$
D31 $\varepsilon^* = 0.035,\ \delta^* = 0.300$
$\delta_{\min} = 0.131$, ATE $= +0.173$, $|\text{bias}|/\delta^* = 0.71$
M32 $\varepsilon^* = 0.110,\ \delta^* = 0.150$
$\varepsilon_{\max} = 0.186$, ATE $= +0.091$, $|\text{bias}|/\delta^* = 0.22$
D32 $\varepsilon^* = 0.070,\ \delta^* = 0.300$
$\delta_{\min} = 0.171$, ATE $= +0.062$, $|\text{bias}|/\delta^* = 0.20$
M33 $\varepsilon^* = 0.110,\ \delta^* = 0.300$
$\varepsilon_{\max} = 0.231$, ATE $= -0.146$, $|\text{bias}|/\delta^* = 0.48$
D33 $\varepsilon^* = 0.110,\ \delta^* = 0.300$
$\delta_{\min} = 0.214$, ATE $= -0.087$, $|\text{bias}|/\delta^* = 0.33$
Probability
ε
δ
Our sharp interval
Assumption-free TP interval
True PoC
Oracle value
TP exogeneity bounds
Compatibility threshold